JAWS: Journal of Arts Writing by Students
Volume 8 Number 1
Issue Art and Non-Human Agencies

© Manon Persoone, 2022. The definitive, peer reviewed and edited version of this article is published in JAWS: Journal of Arts Writing by Students, volume 8, issue Art and Non-Human Agencies, 89-104, 2022, https://doi.org/10.1386/jaws_00044_1.

MANON PERSOONE
KU Leuven and Chalmers University of Technology

The *thigmophiliac* composting stove

ABSTRACT
This investigation into the author's master thesis in architecture starts from an observation that our current ways of coping with waste sustain an obliviousness and feed a lack of responsibility towards the waste we produce and the way it is treated. Moving towards more effective ecological living patterns, but deviating from a strong adherence to technology, raises the question of how architecture can trigger a shift in mindsets on how we deal with waste. While ecological thinking and acting in the field of architecture is often focused on materials and their cycles of re-use and durability, the waste this research consciously takes interest in is of a more uncanny kind. Human bodily waste is the waste exchanged by mediation of the architectural element of the toilet. Instead of focusing on a fast and total removal of waste from our home, can architecture apply principles of care for waste as a *vibrant matter*? Aspiring to evoke ecological awareness, a low-tech toilet design is scrutinized in this paper for its potential to perform care from a *thigmophiliac* point of view, as well as for producing compost.

KEYWORDS

waste
toilet
care
vibrant matter
mindset
architectural design
ecological practice

A. FRAMING OF CHALLENGE

> Would it not be wonderful if the despised object, idea, person, or group, the rubbish heaps of society, were to turn out to be its transformers, containing within their collective corpus the seeds of future regeneration? If what goes down can also come up in the world, all of a sudden, like falling up a social cliff. (Wildavsky 1980: 736)

Departing from the proposed geologic epoch of the Anthropocene, in which humanity has become the major force threatening multi-species liveability on earth (Tsing et al. 2017: G1), this research paper in architecture puts a lens on the human as a creative agent engaged in ecological processes. It starts from an observation that our current ways of coping with waste increasingly put a heavy burden on our environment and sustain an avoidance towards the waste we produce and the way it is treated. As architecture is eminently gifted for the creation of new possible worlds, it is its trump to merge spatial and material with social and cultural issues by means of concrete design strategies and projects centred on habitation. In this capacity, architecture affects and thus helps reconfigure the customary sets of attitudes, habits, values, social forms, practices, and material traits which characterize our patterns of living. Aspiring to move towards more effective ecological living patterns, this article investigates how architecture can transform mindsets on waste and its treatment towards an ecological practice. More specifically, it takes an interest in human bodily waste, the waste exchanged between our human bodies and the earth. Between the microbiome ecologies that digest the food in our gut, and the microbial ecologies flourishing in the soils we colonize and exhaust by growing our crops (De Smet et al. 2021: 168), it is the waste exchanged by mediation of the architectural element of the toilet.



Figure 1: Manon Persoone, The toilet space as smallest room inside the home, 2021. Analogue photograph. 10 x 15 cm. © Manon Persoone

In western industrialized countries, the flushing toilet or WC (water closet) is widespread, using the force of six to eight litres of water to transfer our waste underground to water treatment plants and even rivers and the sea. What is commonly known as the smallest room inside the home (fig.1), is much more dominant in the underground due to its endless kilometres of sewers to transfer waste outside of the city. In the context of Flanders specifically (the northern region of Belgium adjacent to the Netherlands),

this wastewater infrastructure comes down to 46 000 km of sewers which equals going around the world and still having 6 000 km left (Verlinden and Vanoppen 2021: 31). In this dense area, 400 000 houses are not connected to the sewers or to a treatment system and are now discharging their untreated waste into nature. But even when houses are connected, the waste produced there still ends up in nature especially when there is an overflow of their capacity. Due to our climate emergency and an unforeseen growth in population since the Flemish sewers were built, sewer overflows are today happening with acceleration (Verlinden and Vanoppen 2021: 40–46). The problem raised here is not just that these systems of centralized wastewater treatment are outdated and unsustainable, but more so that they sustain our obliviousness on the subject and feed a lack of responsibility towards the way the waste we produce is treated.

In light of moving from an industrial to an ecological era, this article focuses on the understudied, yet potentially key, architectural element of the toilet in the search for an ecological shift in mindsets on waste. That the object of the toilet is underestimated and consequently understudied is well expressed by AMO, the research branch of the leading architectural office OMA in Rotterdam (Office for Metropolitan Architecture), renowned for its inquisitive profile in the field of urbanism and architecture. More specifically, in its leading publication *Elements of Architecture* the specific chapter *Toilet* states 'throughout history no architectural treatise has ever declared the toilet as a primordial element of architecture to start thinking from in relation to our present and future worlds' (Koolhaas 2018: 1557). While conceived as underestimated within the field of architecture, the toilet is for this research paper the main subject of inquiry. In support of a transition towards more effective ecological living patterns but deviating from an adherence to fragmented technofixes (Haraway 2016: 3), this article explores a low-tech toilet design by following a *permaculturist* approach: searching for ways to make everyday systems and techniques that embody and embed care for the earth (Holmgren 2017).

The design emerged in response to calls raised by authors such as Donna Haraway, who describes our global task as to make the Anthropocene as short and thin as possible, not by reconciliation or restoration, but by striving for multi-species recuperation (Haraway 2016: 10), and Maria Puig de la Bellacasa, who searches for ways of thinking that engage care in concrete and worldly ecologies (Puig de la Bellacasa 2017: 18). While the exploration of this design starts from the needs of a human body, it contributes to a non-anthropocentric discourse in architecture by actively considering and involving non-human communities. The notion of *care* in this article is specifically twofold, as the design under focus aims to care for the human body as well as give voice to the non-human others present in our waste and soil. In her *Matters of Care,* Puig de la Bellacasa states 'care is everything that *is* done to maintain, continue, and repair *the world* so that *all* can live in it as well as possible' (Tronto cited in Puig de la Bellacasa 2017: 161, original emphasis). Relating to novel ideas on care*,* and by picking up on the debate around the vital role of non-human communities in repairing the world, this paper elaborates on what a toilet – as mundane as it is, yet a highly technological architectural element – can mean in the exploration of our current ecological collapse. The design explored in this paper is titled the *thigmophiliac* composting stove and concerns a speculative toilet design I made for my grandparents as part of my master thesis *On Thigmophiliac Spaces: The Outhouse in Suburban Paradise* (Persoone 2021).

Toilets and *thigmophilia*

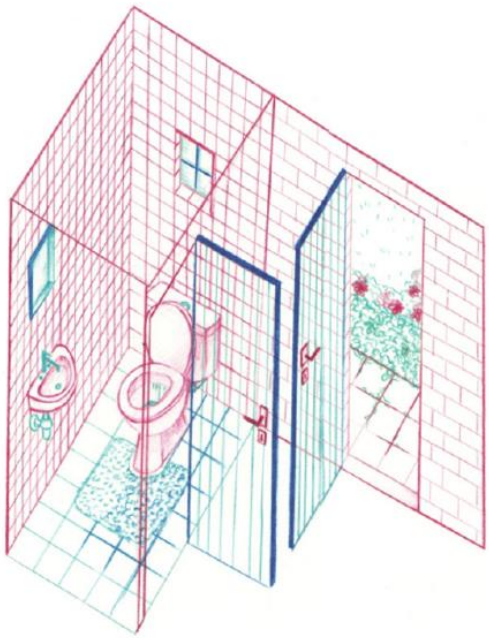


Figure 2: Manon Persoone, Typology of toilet spaces in Flemish Fermettes, 2021.
Drawing in pencil on paper. 29,7 x 21 cm. © Manon Persoone

Coined by biologist Midas Dekkers in *De Thigmofiel: Het verlangen naar geborgenheid* ('The thigmophile: Longing for security'), the term *thigmophilia,* as counterpart to claustrophobia, refers to a secure feeling of comfort caused by being nested and warmed up in a small space (Dekkers 2015: 8). The term originates from the Greek *thigmos* and *philia*, respectively meaning *the touch* and *to love*. Influenced by the information era we find ourselves in today, the sense of sight is dominant in exploring our world. In order to feed our brain of knowledge, we essentially are in need of our eyes. However, *to feel* at ease we need to go back to our primary sense of touch. Even if we are not literally touching, but simply find our bodies' resonance in space, its influence spreads over our whole body (Pallasmaa 2012: 10). When living in a world that is typified by its pressure and a burn-out society, where stress and information overload are part of our daily life, thigmophiliac architecture can care for our bodies and provide us with a valuable moment of rest and solace. Next to the bed as our nest, the toilet is usually also considered a true thigmophiliac space (fig.2) it being the smallest room in the house. When locking ourselves up in this tiny space, we paradoxically create the greatest feeling of rest while relieving ourselves, allowing us to keep the whole world at a distance by simply hanging the occupied sign. The toilet, as the ultimate space to take a moment of rest, is therefore inseparable from the thigmophiliac feeling. Within western industrialized cultures, this private room often has a subordinate role towards the other functions in the home. It hides the object of the toilet between its four walls and is stigmatized by embarrassment because of the unspeakable action happening inside. Where throughout history our conception of bodily waste became increasingly dirty, multiple names were made up in order to keep this natural deed as private and intimate as possible. The toilet as an object and facility has been translated into the washroom, privy, netty, or loo, and the use of a toilet has been cryptically described as going to the restroom, powdering your nose, or relieving yourself. Though it covers one of our most elemental human needs, the overlooked element of the toilet is enclosed and hidden away in our architecture as well as in our use of language.

Can architecture turn our attitude towards the subject of visiting the toilet into performing a valuable act of care? Instead of focusing on a fast and total removal of waste from our home, can architecture apply principles of care for our waste as *vibrant matter*?

Jane Bennet introduces in her *Vibrant Matter* a vital materiality which can be found in matter and things, as opposed to their usual consideration as inert or passive stuff (Bennet 2010: vii). Instead, she

encourages us to engage with the *vibrant* capacity of inanimate things – as waste – to act, to trigger, to impede, to block the will, or to produce effects dramatic and subtle, and see how that could change analyses of political events (Bennet 2010: viii). Likewise, by describing a vitality that exists even in trash, 'Sullivan reminds us that a vital materiality can never really be thrown *away*, for it continues its activities even as a discarded or unwanted commodity' (Bennet 2010: 6, original emphasis).



Figure 3: Manon Persoone, The outhouse (left) and home (right) of my grandparents, 2021. Analogue photograph. 10 x 15 cm. © Manon Persoone

As part of my master thesis on speculative design (Persoone 2021), the *thigmophiliac* composting stove is a toilet which I designed for my grandparents and was fictitiously located in the outhouse of their garden (fig.3). Being aware of my grandparents' love, care, and pride for their garden as a suburban paradise hidden in a residential neighbourhood, I started my design by interviewing them on the beauty of their home and toilet experiences. Having reached age 82 and 78, going outside to use a toilet was a crucial part of my grandparents' childhood experiences. When they were growing up, toilets were isolated from the main body of the house, forming a miniature architectural typology of its own: the outhouse. These isolated huts, characterized by displaying a small heart on its door and poor or non-existing sewage systems, were known to be a bearer of disease. Back then, these houses contained just a simple hole in the ground, a wooden plank on top as a cover, and newspapers to wipe, without any further form of composting or treatment involved. This way, the naturally present viruses, pathogens, and bacteria in bodily waste could not be effectively removed, resulting in the death of outhouses after the sewers network came to prominence.

As opposed to understanding the outhouse in terms of dirt and illness, in Japanese tradition, it is known to be a place of beauty and repose:

> The novelist Natsume Sōseki counted his morning trips to the toilet a great pleasure, *a physiological delight* he called it. And surely there could be no better place to savour this pleasure that a Japanese toilet where, surrounded by tranquil walls and finely grained wood, one looks out upon blue skies and green leaves. (Tanizaki 2001: 7, original emphasis)

In indigenous Asian societies among others, the use of dry toilets is appreciated very differently than in western traditions and some of them even use off-grid treatment systems until today (Thompson 2017: 243). Perhaps their distinct awareness of a close connection to nature explains why their research and practices on off-grid waste management are much more elaborated than in western capitalized environments. The fact is that such techniques and practices promoted nowadays in permaculture are not exclusive to the discipline, but they are shared with or borrowed from agroecology, biodynamical agriculture, and indigenous modes of land care (Puig de la Bellacasa 2017: 125).

Soil care



Figure 4: Manon Persoone, Organisational lay-out of my grandparents' garden, 2021. Drawing in pencil on paper. 29,7 x 21 cm. © Manon Persoone

Building on my grandfather's care work in the garden as his daily moment of rest, he explained the organisation and the lay-out of their vegetable garden to me as well as the different steps taking place in a composting process (fig.4). Together with an experienced gardener, who has been caring for my grandfather since his retirement by assisting him in the harder physical labour, my grandfather fondly spends up to three days a week with his hands in the soil. For him, being able to grow crops and take care of the garden is what brings value to his day (fig.5).



Figure 5: Manon Persoone, My grandparents' vegetable garden, 2021. Analogue photograph. 10 x 15 cm. © Manon Persoone

As opposed to my grandfather's love for the practice of gardening, Puig de la Bellacasa describes the dominant human relation with soil as a way to pace its fertility with production demand, inextricably bringing the resource to exhaustion (Puig de la Bellacasa 2017: 23). While soil remains a resource of extraction value for human consumption and a recalcitrant frontier of inquiry for science, in the context of my grandparents' garden, it is valued as an endangered living world in need of urgent ecological care (Puig de la Bellacasa 2017: 170). Aspiring to shift our human kinship to soil from the one-directional relationship of use to the reciprocity of care, the top layer of the earth's crust is here acknowledged as a multi-species community of biota (Puig de la Bellacasa 2017: 189) which is cared for and nurtured by the practice of composting. Composting is the natural process of decomposing organic matter into a nutrient-rich substance for soil and plants. It contains nitrogen, phosphorus, potassium and all the nutrients and micronutrients that feed plants, enabling them to grow; it also balances soil density, improving the soil's ability to better hold water. While compost is able to nurture plants, for certain weed types, insects, or pests the addition of compost means the garden is no longer a hospitable place to live (Doberski 2022).

The *thigmophiliac* composting stove

After interviewing my grandparents, I started the design phase of the project by debunking all common typologies of the toilet in order to completely rethink this object and space. Based on the values that surfaced in the interview – physical accessibility, safety, hygiene, security, and warmth – I devised *Four Functions of Sanitation* as my design toolkit. Each of these functions represent that set of values and is based on a specific need of the human body when urinating or defecating: body support (physical accessibility), waste treatment (safety and hygiene), enclosure (security and warmth), and cleaning of body and hands (safety and hygiene).

1) Body support: *physical accessibility*

According to a study on *Comparison of Straining During Defecation in Three Positions,* the best position to evacuate was proven to be by squatting (Sikirov 2003). Moreover, the common 90° sitting position was even found to be unhealthy for the human body since it takes a significantly longer period. To provide the most comfortable and healthy toilet-position, where the rectum is in line with the anal canal, Sikirov recommends an angle of 35° between the legs and the upper body. Setting off with my grandparents being the two users of the toilet, I activated their bodies in the design process. While getting in a squatting position is not a possibility for them, the design starts from a sitting toilet accompanied by a footstool of 10 cm, gently rising their legs to influence the hip flexion and consequently the recto-anal angle (fig.6).

2) Waste treatment: *safety, hygiene*

By placing the toilet design outside of the home, this *off-grid* toilet does not depend on centralized treatment systems but instead keeps the produced waste a local matter at the outhouse. As aforementioned, this design explores a permaculturist approach and investigates the composting of human bodily waste as an option to engage with care. As stated in his *Humanure Handbook,* Joseph Jenkins explains one part to our head-in-the-sand approach towards composting human waste by the fact that consumers refuse to deal with their own bodily excretions (Jenkins 2019: 6). While our waste is usually considered dirty and an embarrassment, in the practice of permaculture it is treated and acknowledged as a valuable living resource. By applying a permaculturist approach, engaging with the microbial world flourishing in our waste and soil, the *thigmophiliac* composting stove converts human bodily waste into *humanure* for the garden. As coined by Jenkins, *humanure* is an organic resource material rich in soil nutrients which can be converted to humus by means of a composting process (Jenkins 2019: 8).

The toilet design I propose specifically starts from a urine-diverting dry toilet (UDDT), separating the collection and treatment of urine and excreta (fig.7). Not only does a separation of the two create less odour, but it also serves the advantage that urine can easily be diluted into a fertilizer to water the garden or compost pile (WHO 2018). Since this design concerns a dry toilet, instead of using water, it depends on a combination of dried leaves and sawdust as carbon materials to ignite the composting process. To prevent odours from escaping, sawdust is thrown in the toilet both before and after defecation. Not only does this serve as a scent lock, but it also ensures an ideal composition of the compost heap.

As mentioned before, the failure of the outhouses dating from my grandparents' childhood was a consequence of the many diseases that spread there due to a lack of effective waste treatment or composting to remove viruses, pathogens, and bacteria present in bodily waste. The fact that we harbour many harmful organisms in our waste is the main reason why turning it into compost is considered a dangerous business. One way to do it safely is through hot or *thermophilic* composting. This process involves the cultivation of heat-loving bacteria and fungi which destroy disease organisms existing in humanure by creating a hot environment. According to the USA's Environmental Protection Agency, the compost pile should maintain a temperature above 55°C, and no higher than 72°C, to virtually kill pathogens, viruses, and unwanted bacteria in less than one minute (Jenkins 2019: 196-198). However, these temperatures are only achieved at the centre of the pile. Giving these harmful organisms a year to three years rest without a host will certify that the humanure can be hygienically safe to go back into the soil. As the end-product of the composting stove, humus – similarly to compost – increases the soil's capacity to absorb and hold water when broken into the earth (Jenkins 2019: 22).

3) Enclosure: *security and warmth*

Out of all *Four Functions of Sanitation*, the function of enclosure is responsible for securing as well as providing warmth for the human body. It is the ultimate function to create a thigmophiliac moment of rest. Privately enclosed in the outhouse in their garden, I designed the toilet fixture as a masonry stove constructed of hollow terracotta bricks which are known for their capacity to easily transfer heat (De Decker 2015). As the stove extracts heat from the thermophiliac composting process taking place underneath, it would be slightly warmed up by the residual heat. This way, the use of the composting stove is specifically thigmophiliac as it intimately secures my grandparents' bodies by its shape and warms up their bums while taking a moment of rest. Considering that both the human as the thermophilic creatures are in want of heat, the act of care in this design is two-fold: while the thermophilic bacteria are being fed and nurtured by my grandparents' waste and maintenance of the composting process, they in turn warm up the composting stove by reproducing and converting their energy into heating.

4) Cleaning (body and hands): safety and hygiene

Adding to the design decision of a dry composting toilet, recycled toilet paper is used because it is not whitened with chlorine in its production (Solomon 1996). The addition of unadorned toilet paper can even be beneficial for the compost heap while it is able to absorb a possible excess in moisture (Jenkins 2019: 66). To provide my grandparents with proper hand hygiene after having used the toilet, a sink that is connected to the roof is installed. Here, rainwater is collected, filtered, and slightly heated by the compost before it comes running from the tap (fig.6).

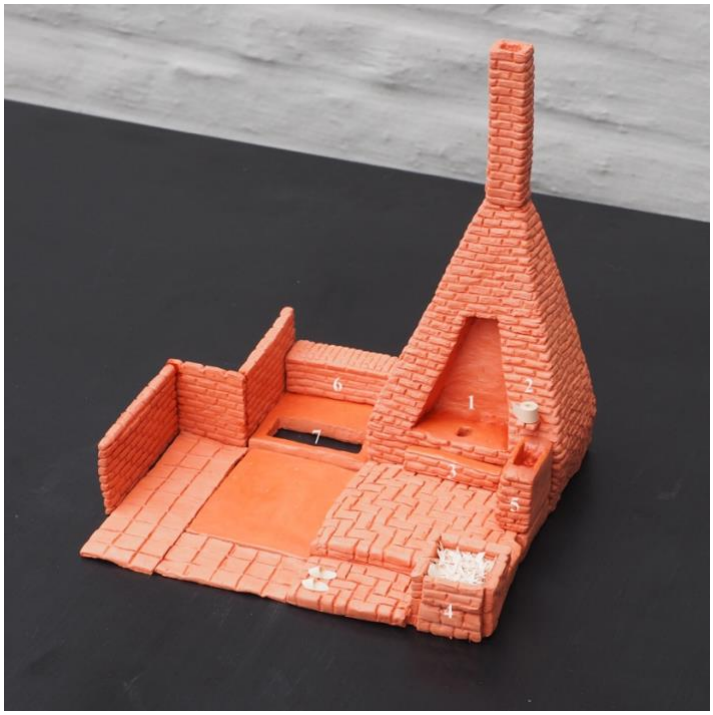


Figure 6: Manon Persoone, Model of the thigmophiliac composting stove with toilet seat (1), toilet roll (2), footstool (3), sawdust bin (4), sink (5), a bench (6), and inspection window (7) to watch the compost underneath, 2021. Photograph. 10 x 15 cm. © Manon Persoone



Figure 7: Manon Persoone, Section through the outhouse and thigmophiliac composting stove, 2021. Drawing in pencil on paper. 29,7 x 21 cm. © Manon Persoone

Spatial organisation of the outhouse

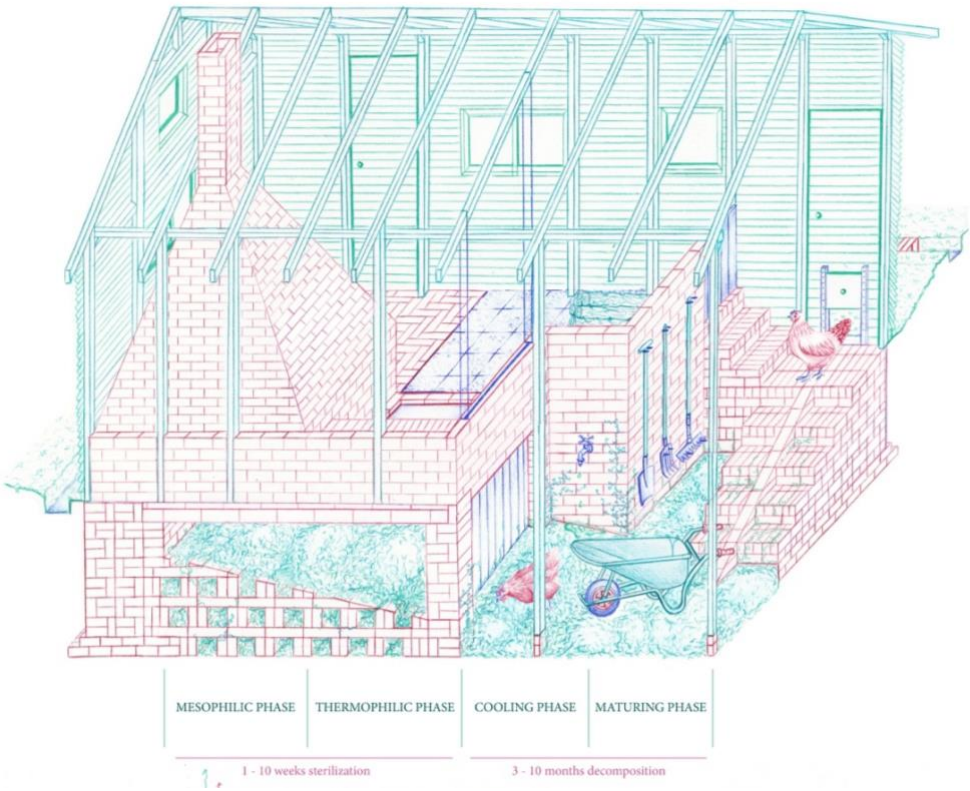


Figure 8: Manon Persoone, Spatial organisation of the outhouse, 2021. Drawing in pencil on paper. 29,7 x 21 cm. © Manon Persoone

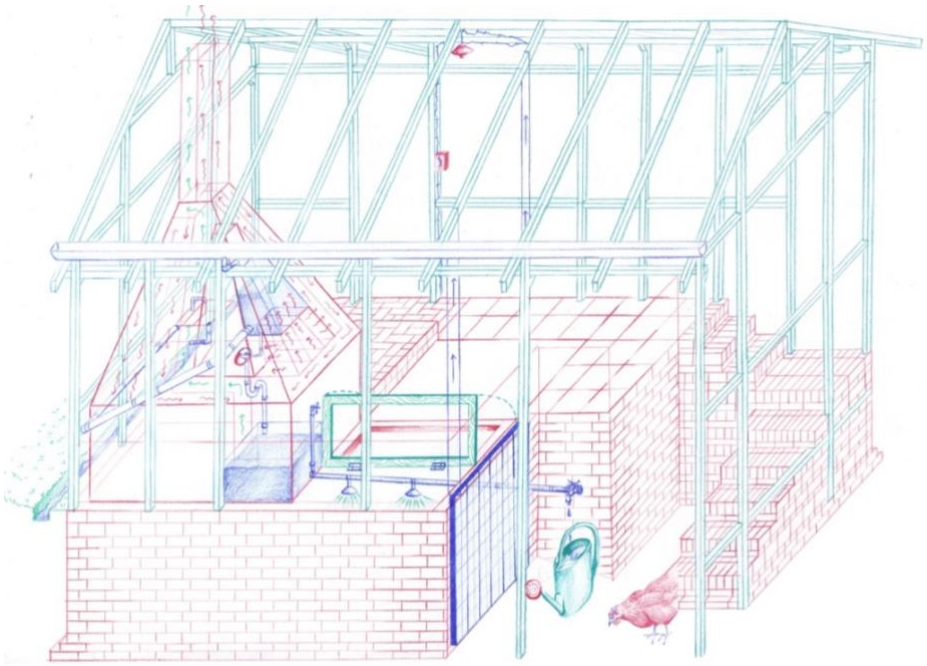


Figure 9: Manon Persoone, Functioning of the thigmophiliac composting stove, 2021. Drawing in pencil on paper. 29,7 x 21 cm. © Manon Persoone

Setting off from the first of its four phases, the first stage in the composting process is called the mesophilic phase (fig.8). In this phase, the bacteria inside of the compost heap combine carbon with oxygen, producing carbon dioxide and energy. While the microorganisms flourishing in the heap use some of the energy for reproduction, they convert most of it into heating (Jenkins 2019: 37). Proliferating and raising the temperature of the entire composting mass, the mesophilic bacteria introduce the second phase of the process, called the thermophilic phase (fig.8).

The thermophilic microorganisms are the most dominant heaters and eaters of the composting process, digesting the coarser organic material during a short but intense heating period. Since the management of this stage is crucial for a safe digestion and conversion of the human waste into humanure, an inspection door and thermometer allow the compost heap to be monitored (fig. 9). In this phase, the compost is in need of more oxygen and moisture as it needs to maintain a temperature above 55°C. Therefore, the collected urine diverted out of the dry toilet and diluted with water can be added to the pile by simply turning a faucet on when the pile gets too dry. At the end of the pipe collecting and transferring urine, there is an additional tap installed to fill a watering can and use the liquid fertilizer directly in the garden (fig.9). Together, the mesophilic and thermophilic phase take about 10 weeks for sterilization of the compost heap (fig.8).

During the third phase, the cooling phase (fig.8), the microorganisms that were chased away by the thermophilic bacteria migrate back into the heap and start to feed on the more resistant organic materials. When arrived at this stage, the compost will have matured into an earthy smell. Since the compost heap is laying on a slope, over time it will gravitationally move towards the closed gate. Having effectively matured into this stage, the heap can safely pass through the safety barrier by opening the gate and helping the heap slide out by using a spade (fig.9). When arrived at this third phase, there is still a lot of food left for the *cool* creatures to flourish in the heap. Since many of the fungi among others, dislike a hot environment, they simply wait for the compost to cool down before digesting, taking them up to ten months to break down the more resistant organic materials (Jenkins 2019: 42).

The final stage of the composting process is the maturing stage (fig.8) and this is a long period, of preferably one year after the thermophilic stage, which serves as a safety net for pathogen destruction. Many of the pathogens present in human bodily waste only have a limited lifespan in the soil, so the longer they are under the effect of this microbiological competition the more likely they will die a swift death. To increase the effect of desiccation, a yearly turning of the pile with a spade can allow more oxygen to enter the pile and fasten the composting process altogether. The most important aspect to the entire composting process is to let the compost reach full maturity in this last phase before taking it out of the outhouse and using it in the garden (Jenkins 2019: 43). The final product that is left after these four phases is a nutrient-rich humus which will improve the soil's structure when broken into the earth.

B. REFLECTIONS

The contribution of this article to the question of 'what a toilet can mean, as a mundane but at the same time highly technological architectural element, in the context of our current ecological collapse' is twofold.

By adopting permaculture techniques into the field of architecture, the thigmophiliac composting stove performs care not only for the human body but also for waste as it becomes *vibrant matter*. The notion of care is present in the toilet's capacity to compost human waste into humanure, engaging with the microbial world flourishing in our waste and soil, as well as in the creation of a thigmophiliac moment of rest for the human user being nested in the masonry stove. Together, they result in an architectural revaluation of using a toilet as well as engaging with waste as a valuable living matter and resource. Building on the creation of a valuable act of care, the design of the thigmophiliac composting stove also expands the dynamics of responsibility towards a form of mutual care. Where the non-human communities present in waste and soil are considered and nurtured by feeding them from our human bodily waste, they in turn provide care for our bodies by removing the pathogens from our waste and producing heat to slightly warm up the composting stove. In a broader context, the design of the *thigmophiliac* composting stove serves as an example of how the material practice of architecture can be turned into a responsible practice of care (Puig de la Bellacasa 2017).

As aforementioned, this contribution also adds to a non-anthropocentric architectural discourse by actively considering and involving the non-human. While there already is a growing body of non-anthropocentric discourse in architectural theory, in western architectural practice this discourse is almost non-existent. Specifically for the architectural element of the toilet, a variety of ecologically inspired designs have already emerged at the intersection of principles of cleanliness and new perspectives on waste but are kept away from exploration on a cultural level while not yet appearing in an urban context. Meanwhile, the use of dry toilets and off-grid waste management has been in use for many years in Indigenous and Asian societies among others, explaining their advanced knowledge in this field. As opposed to our western capitalized environments, these societies do acknowledge human waste as a valuable resource and have been aware of a close connection to nature for many years. In the search for an ecological shift in mindset towards waste and its treatment, western urban environments could take a leaf out of the Traditional Ecological Knowledge book.

Questions

Drawing on the conclusion of my master thesis, my current doctoral research hypothesizes that designing with and for *vibrant matter* would result in fundamentally different spatial settings and in particular, in toilet spaces with different spatial-temporal features. By applying principles of care for waste, the toilet space will have to adopt a radically different position in architecture, not only organisationally and functionally, but also socio-culturally with regards to use, experience, and management of the space. Going into some detail on possible implications, one may think of what it would mean for architecture and our customary culture of inhabitation if human bodily waste would be reassessed and valued.

What would it mean if human bodily waste could not be disposed of as it is now but instead would be kept a local matter to be addressed within the house, landscape, or neighbourhood? What novel spatial practices and new typologies would such an approach of caring for waste as a valuable resource bring to our buildings? Architectural practices that inherently incorporate and literally make room for recovering, cultivating, or even regenerating living matter? Organisations around internal gardens, embedding soils within the house, or layouts that answer to new needs for storage capacity since the production and revaluation of waste is kept an internal matter? And what new collectives could be forged? Collectives that include other species such as microbiota, or even so collectives that explore new forms of neighbouring collaboration?

These are just some of the implications that would change architectural design types and related design strategies, which call for a thorough inquiry within my doctoral research.

C. FORMULATION OF NEW CHALLENGE

In *Flourish: Design Paradigms for our planetary emergency,* Sarah Ichioka and Michael Pawlyn state that carbon offsetting is not the only response to the climate emergency and in fact that 'long-term necessary changes lie primarily in cultural, rather than technological, transformation, and that this must occur at the level of mindsets – the shared ideas that determine to a large extent how society works' (Ichioka and Pawlyn 2021: 2). In light of moving towards more effective ecological living patterns, the aim of this project is to demonstrate ways in which design can transform the mainstream indifferent mindsets on waste, towards a more ecological awareness. As a material practice with the capacity to reconfigure default sets of attitudes, habits, values, social forms, practices and material traits, architecture is foremostly equipped for exploring such a shift but needs a semi-public context when aiming at an effective change in our western culture of habitation. While the design for my grandparents' outhouse was located in a private environment, my current doctoral research focuses on the semi-public living lab *Blikfabriek* in Antwerp (Belgium) as its case study.

ACKNOWLEDGEMENTS

This article has been written within the context of the research line *Architecture & Wicked Matters* and the Academic Design Office (ADO) *The Wicked Home*. Both are part of the KU Leuven Faculty of Architecture's research environment *Material Narratives* and constitute the harbour of this doctoral research. The author would specifically like to thank Prof. Nel Janssens and Dr. Jo(han) Liekens for the discussions leading up to this article.

FUNDING

This article is funded by the KU Leuven Faculty of Architecture.

CONTRIBUTOR DETAILS

Manon Persoone is a Ph.D. student at the KU Leuven (BE) Faculty of Architecture, in double degree with Chalmers University of Technology (SE) Department of Architecture and Civil Engineering. Regarding the research context in Belgium, her Ph.D. is conducted within the research line *Architecture & Wicked Matters* as part of the larger research group *Material Narratives*. In her doctoral research she is currently investigating the understudied, yet potentially key, architectural element of the toilet in the search for an ecologi- cal shift in mindsets on waste and waste treatment. Since 2022, she is lecturer of the master studio titled *Sanitopia* at the KU Leuven Faculty of Architecture. Before obtaining her Master of Architecture, she also gained practical experi- ence while working as a summer-intern at *Robbrecht en Daem architecten*.

Contact: KU Leuven – Faculty of Architecture campus Sint-Lucas, Hoogstraat 51, 9000 Ghent, Belgium.
E-mail: manon.persoone@kuleuven.be

https://orcid.org/0000-0002-4852-1863

REFERENCES

Bennet, Jane (2010), *Vibrant Matter: A Political Ecology of Things*, Durham and London: Duke University Press.

De Decker, Kris (2015), *Stralingsverwarming*: *Gezonde warmte met minder energie* ('Radiant heating: healthy heating with less energy'), Utrecht: Eburon Academic Publishers.

Dekkers, Midas (2015), *De Thigmofiel: Het verlangen naar geborgenheid* ('The thigmophile: Longing for security'), Amsterdam: Atlas Contact.

De Smet, Annelies, Liekens, Jo(han), Janssens, Nel and Persoone, Manon (2021), 'J for Jewel', in M. Fernández Guardado and M. Ballestrem (eds), *CA²RE / CA²RE+ Hamburg: Conference for Artistic and Architectural Research – Book of Proceedings,* Hamburg, Germany, 24-28 March, Hamburg: HafenCity Universität Hamburg, pp. 164-171.

Doberski, Julian (2022), *The Science of Compost: Life, Death and Decay in the Garden,* London: Pimpernel Press Ltd.

Haraway, Donna J. (2016), *Staying with the Trouble: Making Kin in the Chthulucene,* Durham and London: Duke University Press.

Holmgren, David (2017), *Permaculture: Principles and Pathways Beyond Sustainability*, 2nd ed., Victoria: Melliodora Publishing.

Ichioka, Sarah and Pawlyn, Michael (2021), *Flourish: Design paradigms for Our Planetary Emergency*, Axminster: Triarchy Press.

Jenkins, Joseph C. (2019), *The Humanure Handbook: A Guide to Composting Human Manure,* 4th ed., Pennsylvania: Joseph Jenkins, Inc.

Koolhaas, Rem and the Harvard Graduate School of Design (2018), *Elements of Architecture*, Cologne: Taschen GmbH.

Pallasmaa, Juhani (2012), *The Eyes of The Skin: Architecture and the Senses,* 3rd ed., New York: John Wiley & Sons.

Persoone, Manon (2021), 'On Thigmophiliac Spaces: The Outhouse in Suburban Paradise', MA dissertation, Ghent: KU Leuven Faculty of Architecture.

Puig de la Bellacasa, Maria (2017), *Matters of Care: Speculative Ethics in More Than Human Worlds,* Minneapolis: University of Minnesota Press.

Sikirov, Dov (2003), 'Comparison of Straining During Defecation in Three Positions: Results and Implications for Human Health', *Digestive Diseases and Sciences,* 48:1, pp. 1201-1205.

Solomon, K. R. (1996), 'Chlorine in the Bleaching of Pulp and Paper', *Pure and Applied Chemistry* 68:9, pp. 1721-1730.

Tanizaki, Jun'ichirō (2001), *In Praise of Shadows*, rev. ed., London: Vintage Classics

Thompson, Michael (2017), *Rubbish Theory: The Creation and Destruction of Value*, rev. ed., London: Pluto Press.

Tsing, Anna L., Swanson, Heather A., Gan, Elaine and Bubandt, Nils (2017), *Arts of Living on a Damaged Planet: Ghosts and Monsters of the Anthropocene*, Minneapolis: University of Minnesota Press.

Verlinden, Toon and Vanoppen, Marjolein (2021), *Weg van Water: Een reis door België (z)onder water* ('Way of water: A journey through Belgium without or under water'), Ghent: Academia Press.

Wildavsky, Aaron (1980), 'Delving into dustbins', review, *Times Literary Supplement*, pp. 736-736.

World Health Organization (2018), *Guidelines on sanitation and health*, Geneva. Licence: CC BY-NC-SA 3.0 IGO.